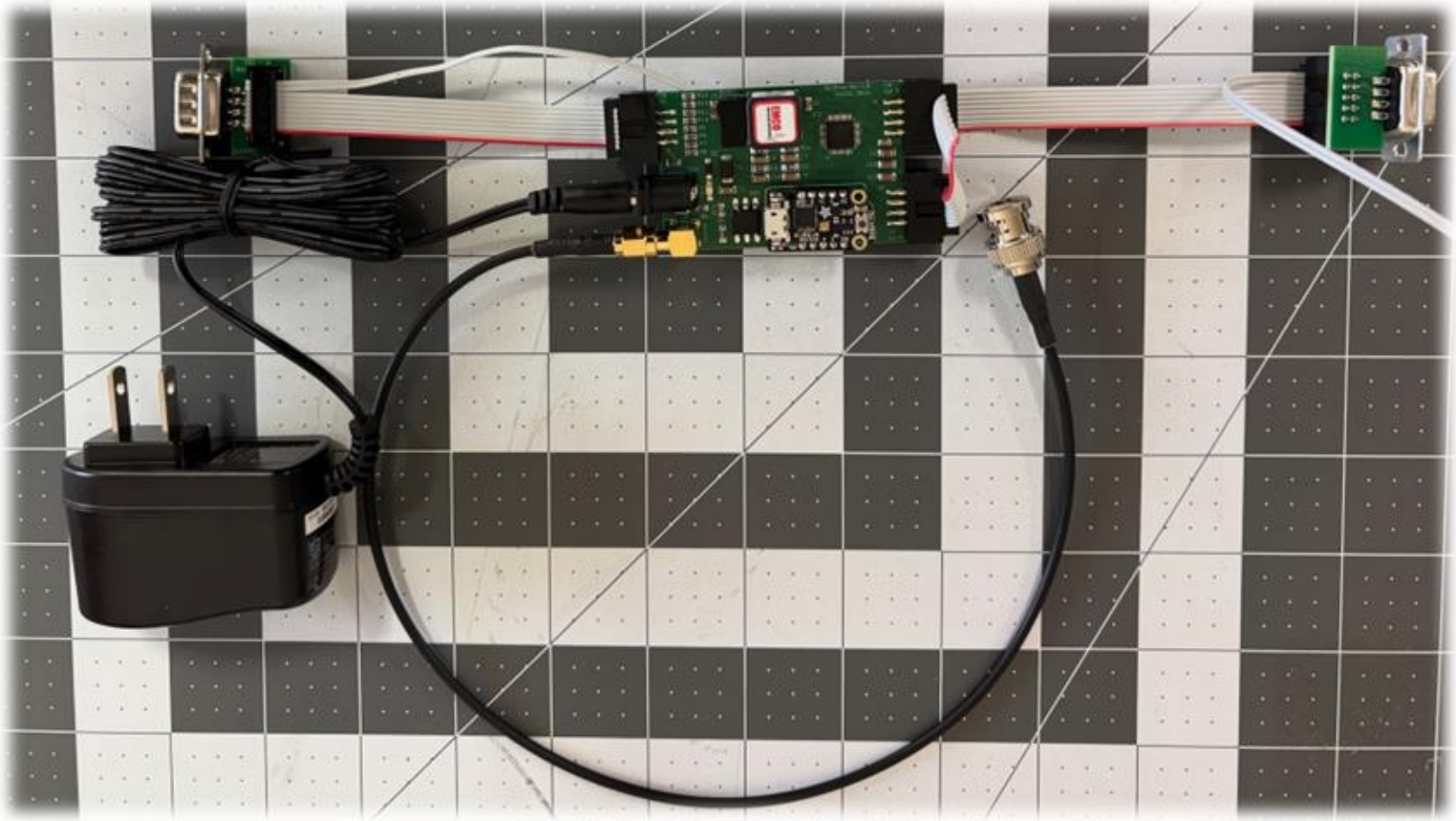
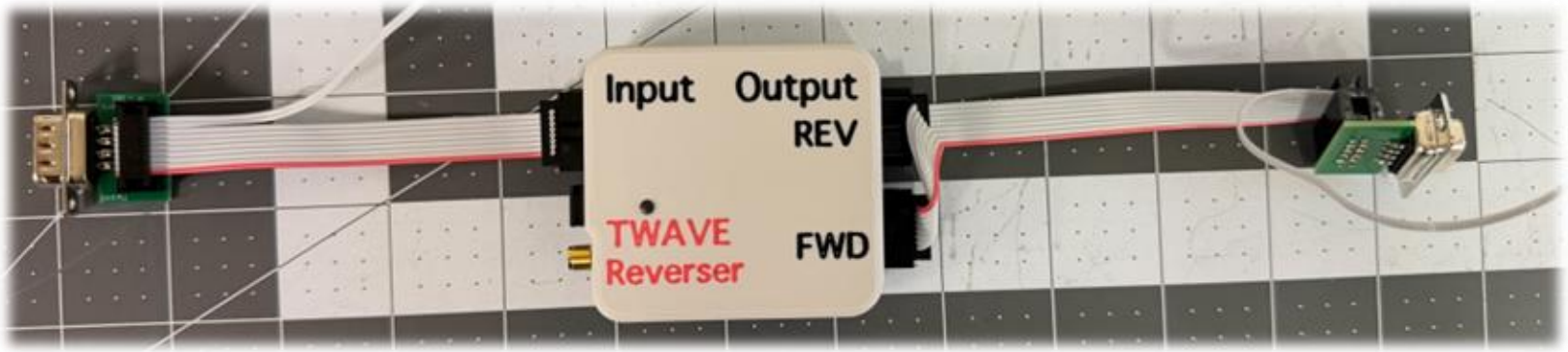
TWAVE reverser (SLIM Reverser), 4.0

GAA Custom Electronics, LLC



TWAVE Reverser in encloser

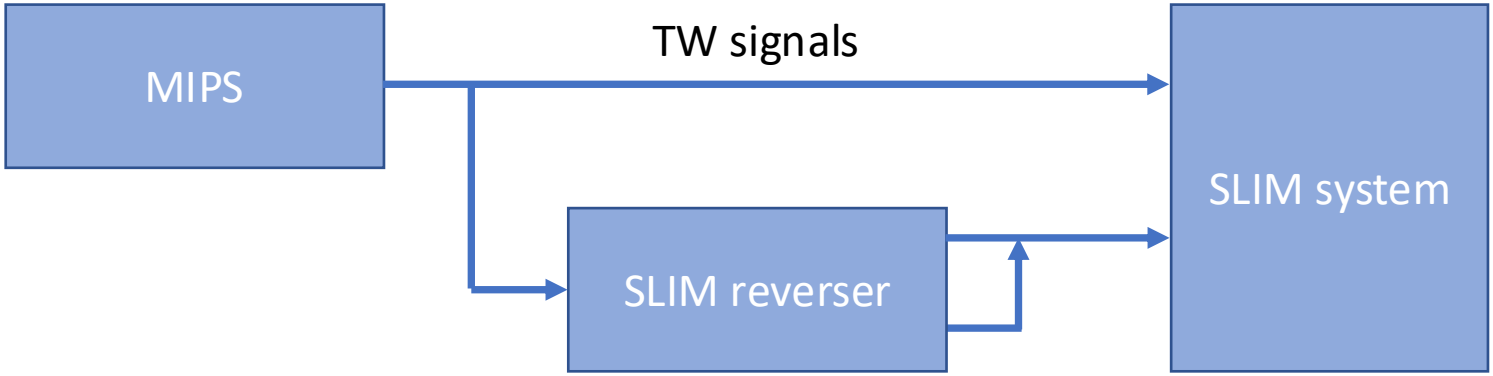


SLIM reverser usage

The SLIM reverser does not generate a TW set of output signals. The switch is designed to pass a set of TW signals thru or reverse the signal order and reverse the direction of the TW. This can be very useful in SLIM designs where you need to control the direction of the ions without the need for an additional set of TW signals.

The SLIM reverser has two sets of outputs, a normal non-reversed output and a reversed output. When one set of outputs are active the other set is high impedance. To make a simple reverser the two output sets are connected together.

In a typical application an existing TW set of outputs are connected to the switch as well as used in your SLIM device. The SLIM reverser outputs are then connected to the SLIM section you wish to reverse.



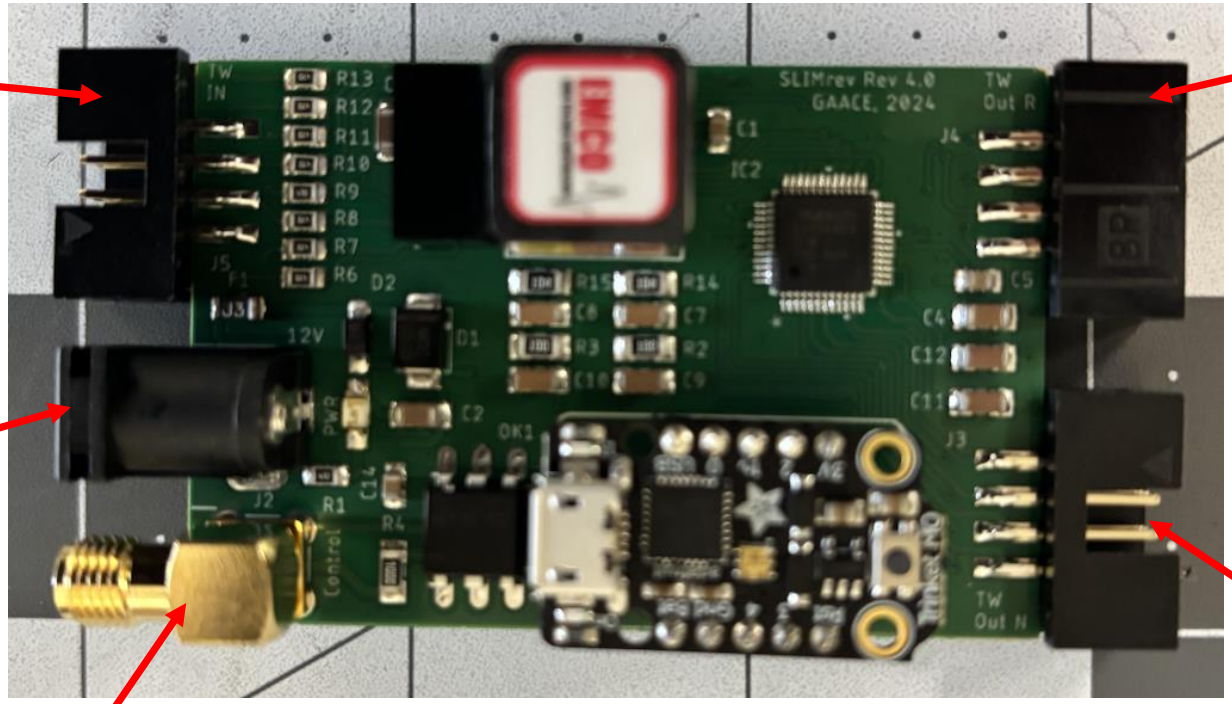
SLIM reverser features

- 16 pole analog bi-directional switch
- Two modes
  - Straight thru, signals are passed through the switch to the normal output connector
  - Reverse, TWAVE signal order is reversed and sent to the reverse output connector
- 100Vp–p maximum differential voltage between TW signals
- Signal shapes are preserved through the switch, sine, pulse, triangle, etc
- Signal amplitudes are preserved thru the switch
- Input TW can be floated
- Insolated control input
- Input control 0 to 5 volts
  - 0 volts = straight thru
  - 5 volts = reverse
- Powered from 12VDC wall supply
  - Apply power to switch before applying TW signals

SLIM reverser connections

TWAVE input
8 Pin IDC
Pin 1 – TW 1
Pin 2 – TW 2
Pin 3 – TW 3
Pin 4 – TW 4
Pin 5 – TW 5
Pin 6 – TW 6
Pin 7 – TW 7
Pin 8 – TW 8

TWAVE out R
8 Pin IDC
Pin 1 – TW 8
Pin 2 – TW 7
Pin 3 – TW 6
Pin 4 – TW 5
Pin 5 – TW 4
Pin 6 – TW 3
Pin 7 – TW 2
Pin 8 – TW 1



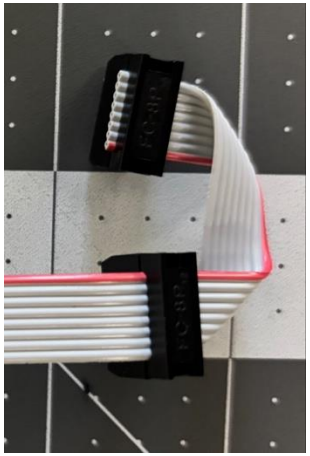
+12VDC power

TWAVE out N
8 Pin IDC
Pin 1 – TW 1
Pin 2 – TW 2
Pin 3 – TW 3
Pin 4 – TW 4
Pin 5 – TW 5
Pin 6 – TW 6
Pin 7 – TW 7
Pin 8 – TW 8

SMA control Input, 0 to 5V, isolated

The switch signals are bi-directional either side of the switch can be inputs or outputs

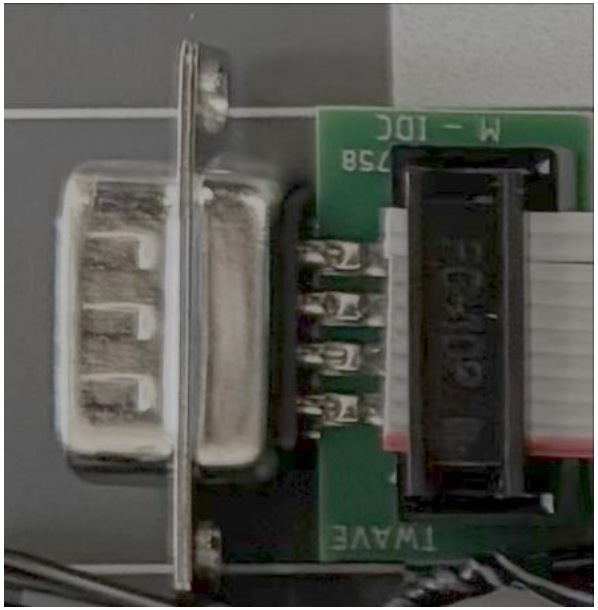
Configuration options



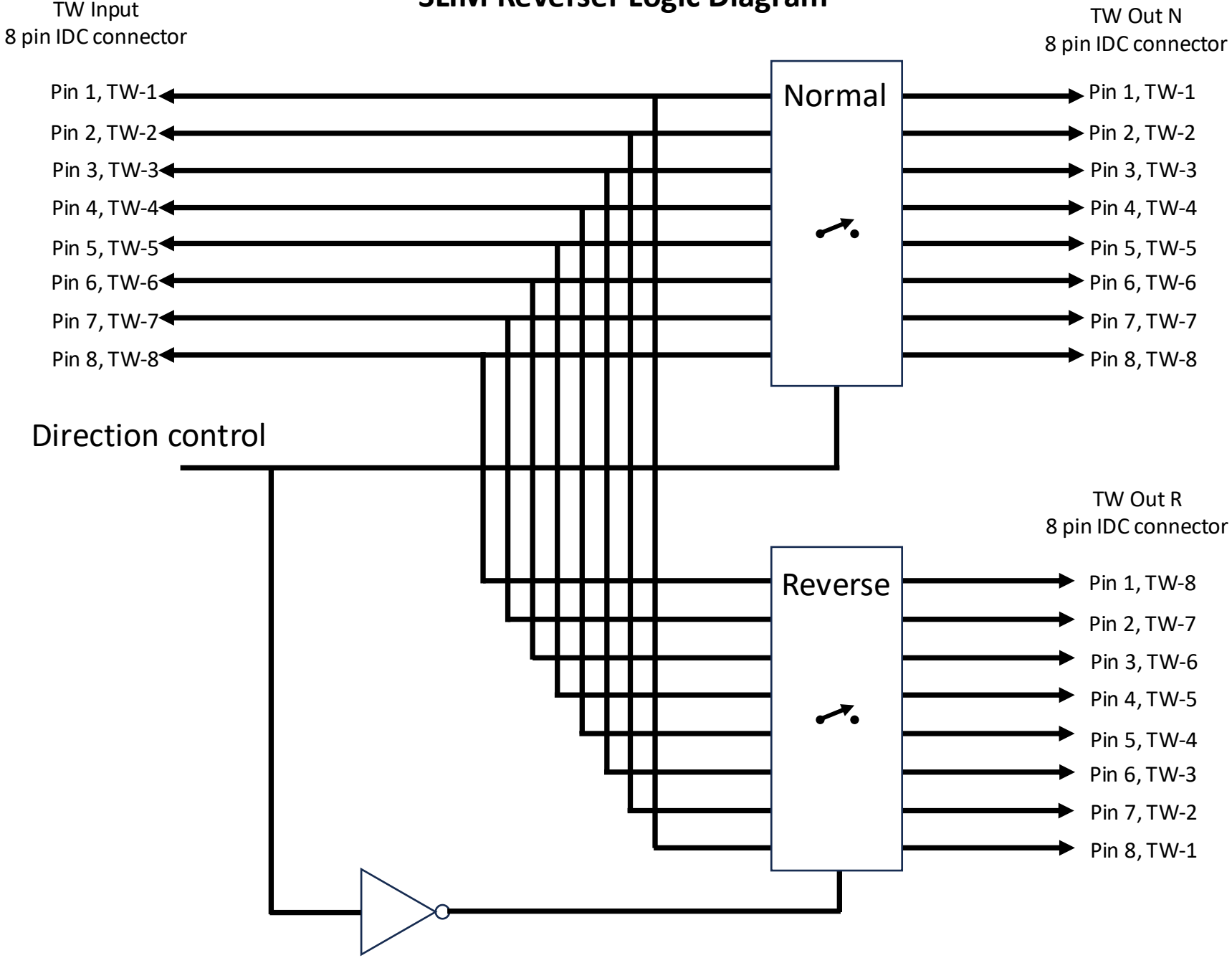
- The reverser output has a normal (non-reversed) output connector and a reversed output connector. When one set of output are active the other set are high impedance. To enable reverse function these two outputs are tied together. This can be done with an IDC crimp cable as show.
- The connectors allow you to generate unique reversal configurations supporting your specific system needs. As an example; if the normal connector is used with conductor pins 1 thru 8 crimped to connector pins 1 thru 8 and the reverse connector swaps conductors 1 thru 4 with 5 thru 8 the reversal will be: 1-8,2-7,3-6,4-5,5-4,6-3,7-2,8-1. Any desired output can be achieved by select the proper conductor to connector pin assignment. Flat cable conductors can easily be separated and reposition in the crimp connector before crimping to the cable.

9 pin D to 10 pin IDC adapters

9 Pin D
Pin 1 – TW 1
Pin 2 – TW 2
Pin 3 – TW 3
Pin 4 – TW 4
Pin 5 – TW 5
Pin 6 – TW 6
Pin 7 – TW 7
Pin 8 – TW 8
Pin 9 - Common



10 pin IDC
Pin 1 – TW 1
Pin 2 – TW 2
Pin 3 – TW 3
Pin 4 – TW 4
Pin 5 – TW 5
Pin 6 – TW 6
Pin 7 – TW 7
Pin 8 – TW 8
Pin 9 and 10- Common

Both 9 pin male and female D connector to 10 pin IDC adapters are provided. These adapters will allow you to connect to the 8-conductor cable from the reverser connectors. The 10 pin IDC also provides two common or ground ,conductors, 9 and 10. The Reverser does not require the ground conductors and establishes a reference to the incoming TW signals.

SLIM Reverser Logic Diagram